2025 - IS6611 - IT Artefact V3

Predicting Business Star Ratings from Early Review Text Using NLP

Group 6:

Aida F. George	**124109805**
Aiyman Mohammed	**124110673**
Aravind Ravichandran	**124109178**
Dheeraj Ravi	**124114671**
Mrunal Howale	**124112093**
Prashanth Punniyaseelan	**121108055**

Table of Contents

Table of Figures

1. Executive Summary

Figure 1: Yelp

In today's digital economy, online reviews significantly shape consumer perception, especially in the food service industry, where early customer feedback can define a business's reputation (BrightLocal, 2020; Luca, 2016). However, platforms like Yelp often reflect customer sentiment retrospectively, resulting in lost opportunities for early intervention. This project proposes a proactive machine learning solution that predicts long-term business ratings using only the early or recent few customer reviews, enabling data-informed decision-making before reputation damage occurs.

Using the Yelp Open Dataset (Yelp, 2023), we conducted extensive data cleaning, preprocessing, and exploratory analysis to focus on mid-sized food establishments. From this refined dataset, we engineered features that capture sentiment (via VADER; Hutto & Gilbert, 2014), review engagement metrics (votes, adjectives), temporal behaviour, and potential fake review signals. A combination of TF-IDF vectorization and structured features was used to train multiple classification models, including Logistic Regression, Random Forest, and

XGBoost. Among these, XGBoost demonstrated the best performance with an overall accuracy of 61% and strong interpretability.

Insights are delivered through a user-friendly Power BI dashboard, equipped with SHAP-based feature explanations and a 'What-If' input simulator (Lundberg & Lee, 2017). The solution directly supports Sustainable Development Goal 8 (Decent Work and Economic Growth) and SDG 9 (Industry, Innovation and Infrastructure) by embedding advanced analytics into real-world SME operations (United Nations, n.d.).

Our artefact bridges the gap between unstructured customer feedback and actionable insight, offering a scalable framework for predictive reputation management in the food industry.

2. Problem Definition and SDG Alignment

2.1 Problem Statement

Customer reviews on platforms like Yelp are crucial to the reputation and success of small and mid-sized food businesses (SMEs) (Luca, 2016). However, these reviews are retrospective, often alerting owners to dissatisfaction only after reputational damage has occurred (BrightLocal, 2020). Since the earliest reviews disproportionately shape public perception and visibility, delayed responses limit opportunities for timely intervention and service improvement (Mishra & Rastogi, 2019).

This project addresses the issue by using Natural Language Processing (NLP) and machine learning to predict a business's long-term average rating based solely on its first or the recent 3 to 5 reviews (Khalid et al., 2023). By transforming review monitoring from a passive to a proactive process, the artefact empowers business owners to identify early warning signs and act before negative sentiment escalates.

2.2 Process Transformed

This data-driven approach enables timely interventions by highlighting drivers of customer dissatisfaction through interpretable SHAP insights (Lundberg & Lee, 2017). The process shifts from reactive observation to proactive reputation management, supporting sustained competitiveness and customer retention.

2.3 Sustainable Development Goals Alignment

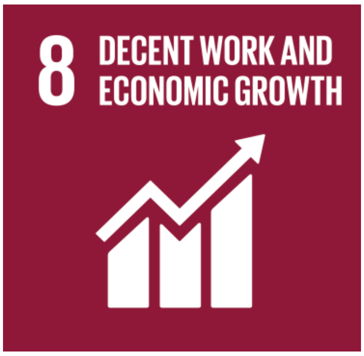


Figure 2: SDG 8

SDG 8 – Decent Work and Economic Growth
It helps SMEs enhance service quality, retain customers, and protect local employment by enabling early action on customer feedback (United Nations, n.d.).

Figure 3: SDG 9

SDG 9 – Industry, Innovation and Infrastructure
By embedding AI-driven analytics and explainability tools into SME workflows, it democratizes access to innovation and fosters sustainable technological adoption (United Nations, n.d.).

In uniting predictive intelligence with real-world business needs, the solution builds operational resilience and contributes meaningfully to inclusive economic growth.

Current Review Monitoring Process (AS-IS)

	Start

Customer	**Writes a review after visiting a business.**
Yelp	**Review is posted publicly on the review platform (e.g., Yelp), contributing to the business's aggregate star rating.**
Business Analysis/Systems	**No action is taken immediately. Multiple reviews accumulate over time.**
Owners/ Analysts	**Business owner or analyst manually reads and interprets trends, often reactively, based on negative reviews or falling ratings.**
Action	**By the time action is taken (e.g., service change, response strategy), the reputation has already suffered, and customer churn may have started.**
	End

Proposed Predictive System (TO-BE)

	Start
Customer	**Customer submits a review.**
Yelp integrated system	**Review is captured by an integrated pipeline and automatically sent to the NLP engine.**

Business Analysis/Systems	**NLP model processes review and generates sentiment score, behavioral indicators, and a predicted long-term star rating.**
Dashboard	**Dashboard displays the predicted outcome along with feature-level explanations (via SHAP) to the business owner.**
Alert/Signal	**The system flags at-risk businesses (e.g., sharp drop in predicted rating, negative sentiment trend).**
Action	**Business owner takes proactive action and adjusts operations, responds to reviews, initiates customer outreach, before public perception declines.**
	End

3. Data Value Map

This project draws on the Yelp Open Dataset, a publicly available and widely cited benchmark in review analytics, to model early customer sentiment and long-term business performance. Specifically, we used two core files: *review.json*, containing over 8 million anonymized reviews with timestamps, ratings, and vote metrics; and *business.json*, which provides metadata such as business location, categories, and operating hours.

To ensure analytical focus and data quality, we scoped the project to food-related small and mid-sized enterprises (SMEs). We applied several filtering steps: retaining only food-service categories (e.g., restaurants, cafes, bakeries), excluding the top 100 most-reviewed businesses (typically large chains), and selecting businesses with 20 to 750 reviews to avoid skew from outliers (Mishra & Rastogi, 2019). Only reviews from 2015 onwards were included to reflect recent sentiment patterns (BrightLocal, 2020). We also excluded users with fewer than three reviews to reduce spam risk. This process resulted in a curated dataset of approximately 1.93 million reviews across diverse U.S. regions.

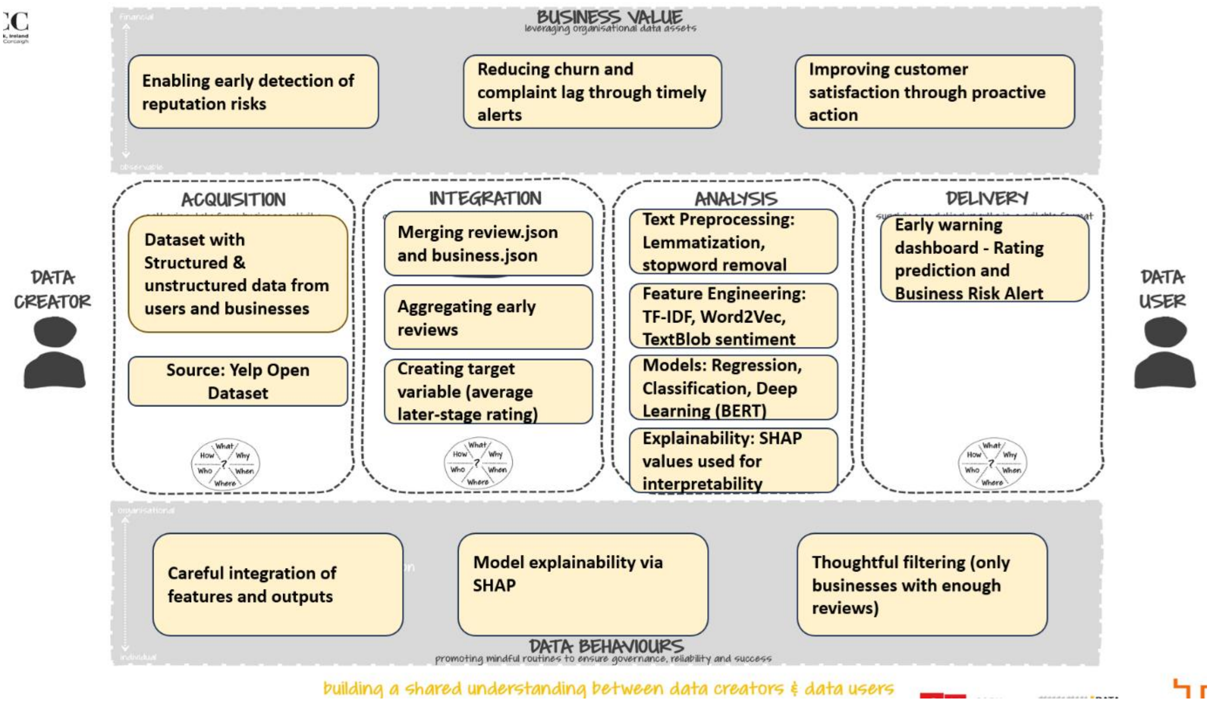


Figure 4: Data Value Map

3.1 Ethical and Quality Considerations

Ethical and quality considerations guided the acquisition process. The dataset is anonymized and freely accessible under Yelp's terms, eliminating privacy risks (Yelp, 2023). To address concerns around fake reviews, we engineered features to flag suspicious patterns such as mass posting across businesses, repeated phrases, and bursts of reviews in a short timeframe (Khalid et al., 2023). These heuristics were designed to identify low-quality or manipulated entries and improve model robustness (BrightLocal, 2020).

Originally stored in nested JSON format, the data was parsed and transformed into structured CSVs using Python. The final dataset, drawn from an initial pool of over 6.7 million entries, is both reliable and research-ready, supported by its consistent schema, peer-reviewed usage, and open documentation (Sammon, 2024; Yelp, 2023). This made it a suitable and trusted foundation for our artefact.

3.2 Integration

Integration involved transforming and unifying the Yelp dataset into a structured, analysis-ready format. Using business_id as the primary key, we merged *review.json* and *business.json* via an inner join to retain only reviews linked to businesses with complete metadata (e.g., categories, location, hours). This ensured consistency and preserved relational context for modelling (Mishra & Rastogi, 2019).

To simulate early customer feedback conditions, we filtered for only the first 3 to 5 chronological reviews per business. We then computed a target variable: the average star rating from all subsequent reviews, which served as the ground truth for the classification task (Khalid et al., 2023). This setup allowed us to predict long-term outcomes based solely on initial sentiment, a key innovation of the artefact.

We also standardized Yelp's unstructured multi-label category system. A keyword-mapping function was created to group over 75,000 combinations into 16 generalized categories (e.g., "Bakery," "Mexican Restaurant"), supporting effective sampling and stratified analysis.

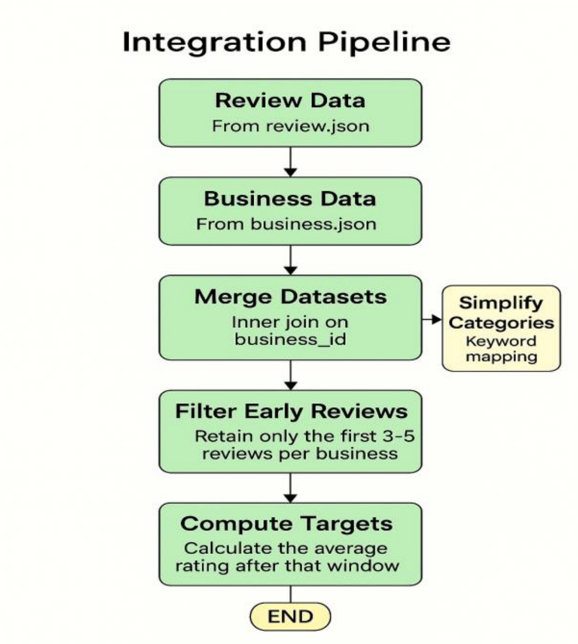


Figure 5: Integration Pipeline

Further refinement steps included removing the top 100 most-reviewed chains, excluding businesses with fewer than 20 or more than 750 reviews, and dropping incomplete records. Reviews under five words or from users with fewer than three posts were excluded to reduce noise (Mishra & Rastogi, 2019; Khalid et al., 2023). Timestamps were converted into datetime objects to support post-2015 filtering and temporal analysis (BrightLocal, 2020).

Technical challenges, such as nested attribute structures and irregular business hour formatting, were handled using ast.literal_eval, pandas.Series(), and custom parsers. Categorical fields were type-cast and label-encoded for machine compatibility. Threshold-based filtering removed rows with insufficient metadata, enhancing reliability.

The resulting dataset, reduced from 6.7 million to roughly 1.93 million high-quality entries across 65 structured columns, served as the backbone for all downstream tasks. Its construction aligns fully with the Data Value Map (DVM) and ensures both analytical depth and operational readiness.

3.1 Data Analysis

3.3.1 Exploratory Data Analysis (EDA)

To build a strong foundation for modelling, we performed extensive Exploratory Data Analysis (EDA) on the filtered dataset of ~1.93 million food-related business reviews. Univariate analysis of the stars variable revealed a positively skewed distribution, with most businesses clustered between 3.5 and 4.5 stars, and relatively few rated below 2.0 — indicating a generally favourable sentiment across Yelp reviews (Luca, 2016; BrightLocal, 2020).

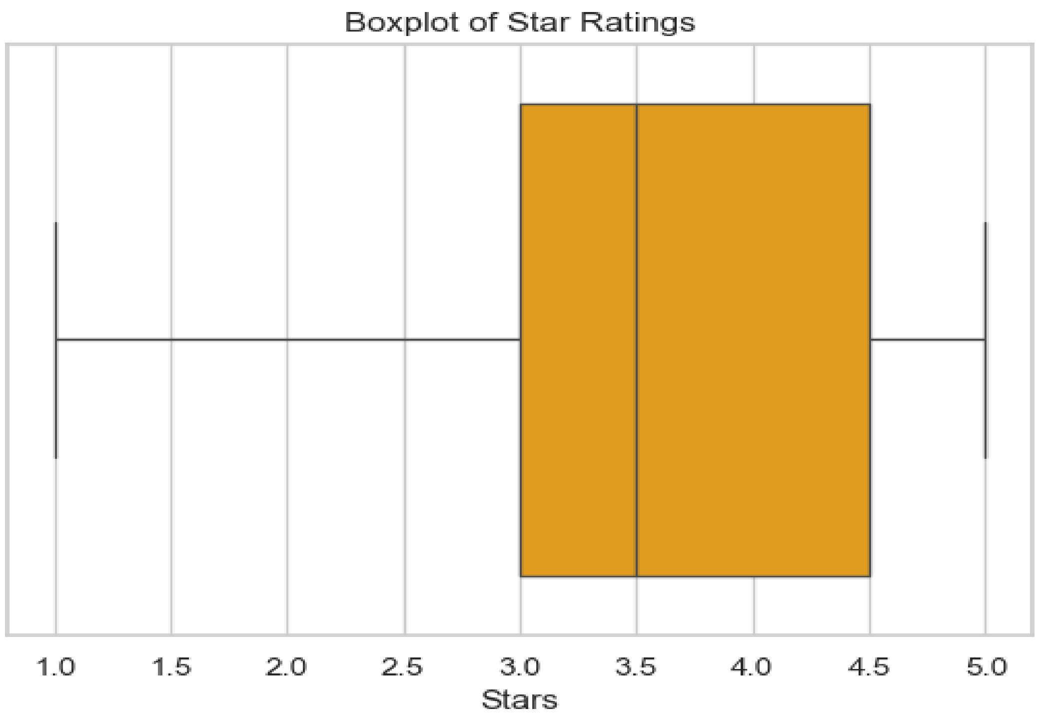


Figure 6: Box Plot of Star Ratings

A histogram of review counts, plotted on a log scale, uncovered a heavy long-tail distribution. While many businesses had under 100 reviews, a small subset received thousands, consistent with platform usage asymmetry.

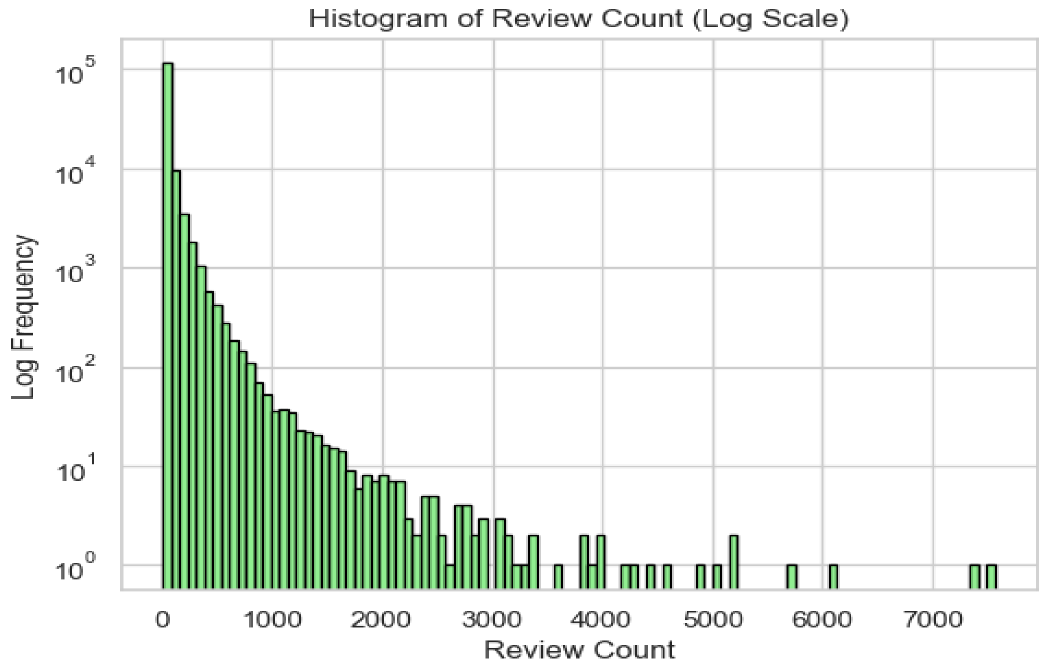


Figure 7: Histogram of Review Count

Boxplots showed clear variation in star ratings across business categories, with bookstores, Asian restaurants, and steakhouses typically receiving higher ratings than fast food outlets or automotive services.

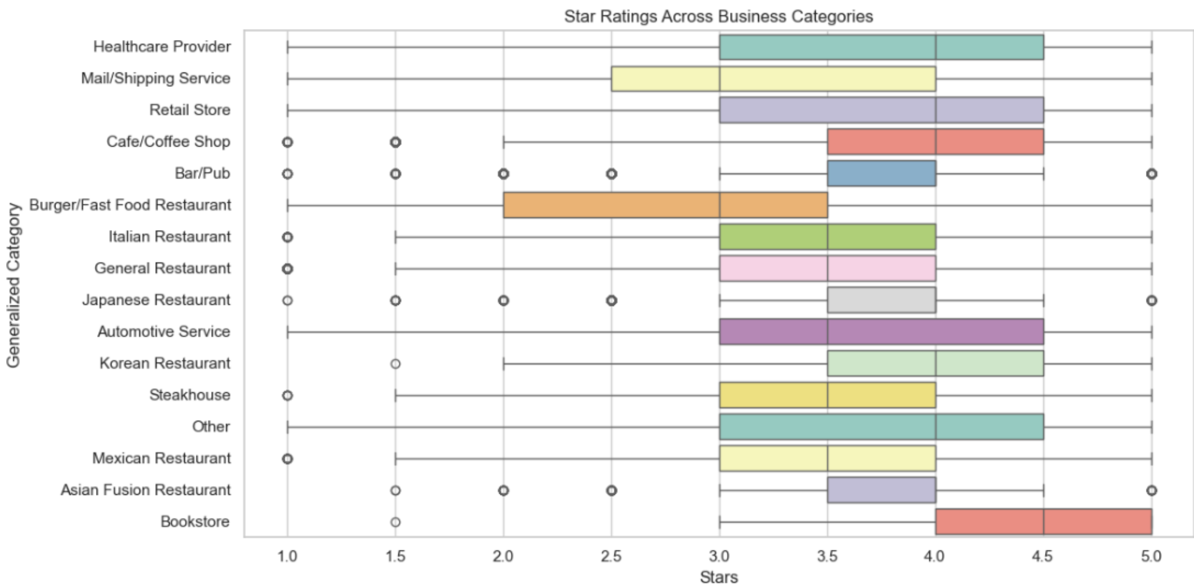


Figure 8: Box Plot of Star Ratings Across Business Categories

Geographic density plots using KDEs for latitude and longitude revealed regional business clustering, most notably around U.S. urban centres. Finally, bivariate plots demonstrated weak or no linear correlations between star ratings and operational attributes like total weekly hours or review volume, reinforcing the need for more complex, non-linear modelling approaches. These findings informed our feature selection, data transformation, and classification strategy.

3.3.2 Prediction modelling

The analysis phase aimed to predict a business's long-term average star rating using only its first few reviews. This was achieved through a multi-step pipeline combining Natural Language Processing (NLP), structured feature engineering, and supervised machine learning, with a strong emphasis on model interpretability (Khalid et al., 2023; Mishra & Rastogi, 2019).

Text preprocessing included lowercasing, punctuation removal, stopword filtering, tokenization, and lemmatization. We then extracted review length, sentiment polarity using VADER (Hutto & Gilbert, 2014), and the frequency of adjectives and adverbs using part-of-speech tagging metrics designed to capture both emotional tone and linguistic complexity (TextBlob Developers, n.d.).

To detect low-quality or synthetic reviews, we introduced behavioral features such as user repetition across businesses (indicative of spamming), repeated phrases (copy-paste patterns), and review bursts posted in short windows. These indicators helped mitigate review manipulation and improved model reliability (BrightLocal, 2020; Khalid et al., 2023).

We transformed reviews into numerical vectors using TF-IDF and retained the top 100 features. These were combined with structured inputs like sentiment and metadata to form the final model-ready dataset. For classification, we discretized the long-term average rating by rounding it to the nearest whole number (1 to 5), enabling a multiclass setup while preserving ordinal relationships (Luca, 2016).

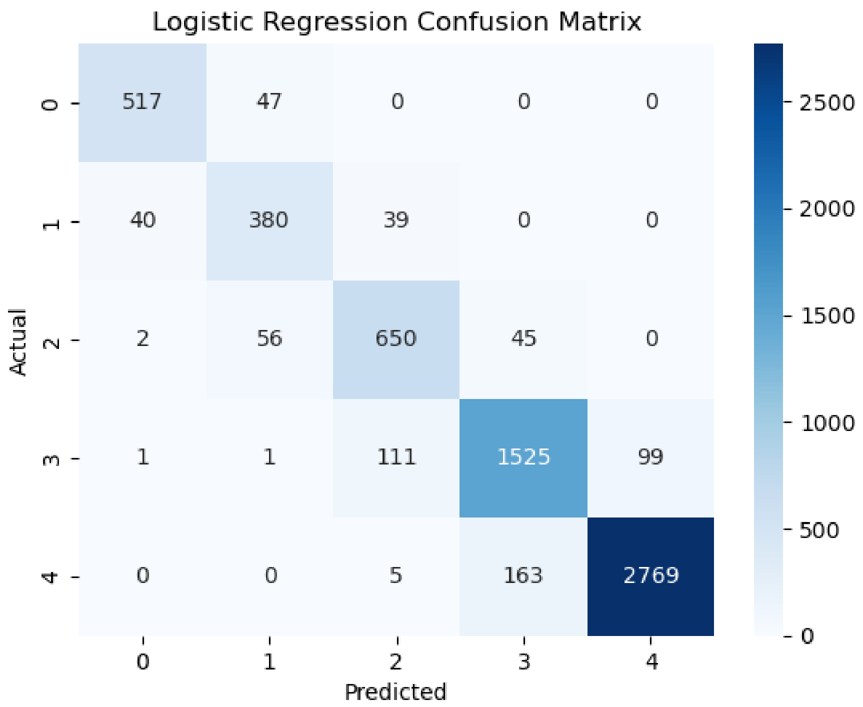


Figure 9: Logistic Regression Confusion Matrix

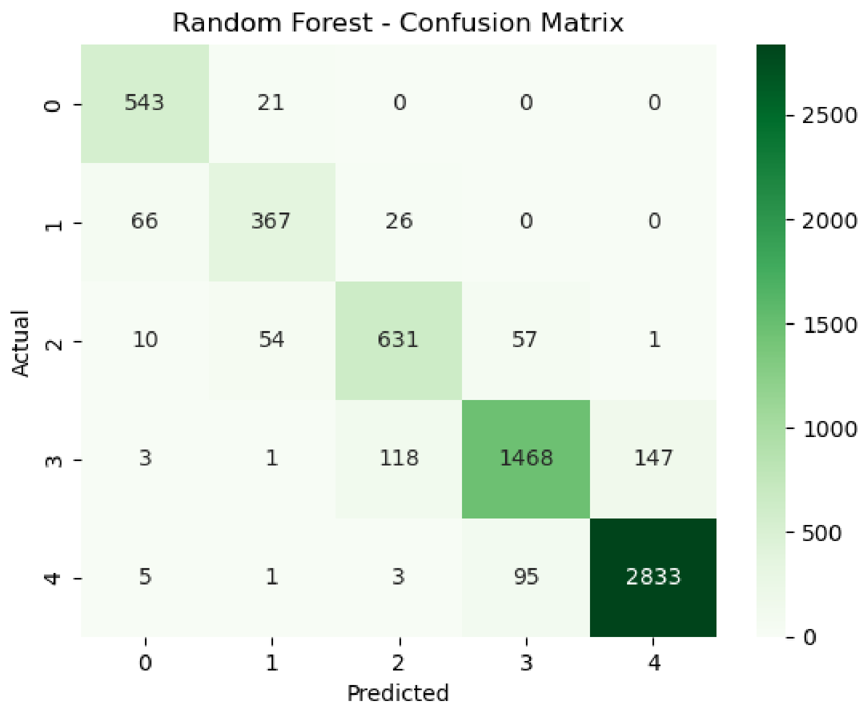


Figure 10: Random Forest - Confusion Matrix

Three classifiers were tested: Logistic Regression (baseline), Random Forest, and XGBoost. Logistic Regression achieved 56% accuracy, excelling at polarized ratings. Random Forest improved generalization slightly, reaching 57%. XGBoost delivered the best performance with

61% accuracy, offering a superior balance across precision, recall, and F1 scores. Due to its interpretability and strong performance on mixed feature types, XGBoost was chosen for deployment.

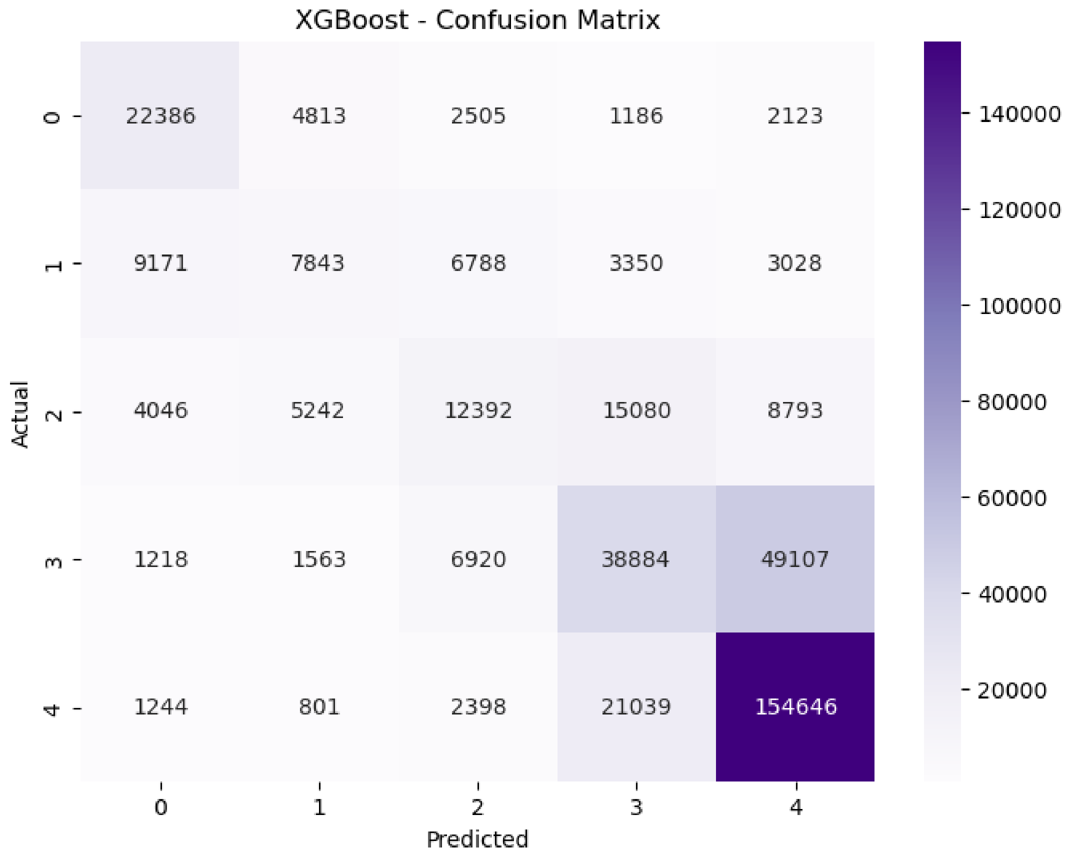


Figure 11: XGBoost Confusion Matrix

Model transparency was achieved using SHAP (SHapley Additive exPlanations), which assigns feature-level contributions for individual predictions (Lundberg & Lee, 2017). SHAP analysis highlighted sentiment score, review length, useful/funny votes, and expressive language as key drivers of predicted ratings. Location-based attributes like city and state also played a role, reflecting regional variation in customer sentiment (Khalid et al., 2023). These insights were later embedded into the dashboard, allowing users to explore not just model outputs but the rationale behind them.

This integrated pipeline spanning data cleaning, model development, and SHAP-based interpretation formed the analytical core of the artefact. It enabled a transparent, high-accuracy solution suited to practical business decision-making.

3.3.3 Why Classification Instead of Regression?

Although average ratings might appear continuous, Yelp only allows users to rate in fixed intervals (1.0, 1.5, 2.0 … up to 5.0). Since these outcomes are discrete and ordinal, we framed the problem as a multi-class classification task rather than regression. This allowed us to predict the most likely star category, simplify interpretation, and improve alignment with how users and business owners perceive ratings in real-world contexts.

3.4 Delivery

The delivery phase focused on translating predictive insights into an interactive, user-friendly Power BI dashboard designed for small and mid-sized food business owners. This interface enables real-time exploration of model outputs, supports data-informed decisions, and fosters trust through built-in interpretability tools like SHAP explanations (Lundberg & Lee, 2017) and a What-If simulator.

To align with decision makers' needs, we focused on three priorities: (1) Early detection of reputation risk before public star ratings drop, (2) Interpretability of predictions to build trust in model outputs, and (3) Usability for non-technical stakeholders such as small business owners and marketing managers (United Nations, n.d.). These needs informed our design with color-coded KPIs, simple risk flags, and embedded tooltips, ensuring stakeholders can extract insights and take action with minimal training.

At its core, the dashboard displays XGBoost-predicted long-term star ratings alongside actual historical averages. This allows users to assess whether early reviews indicate potential underperformance or stable reputation, prompting timely intervention before public ratings are impacted (Khalid et al., 2023).

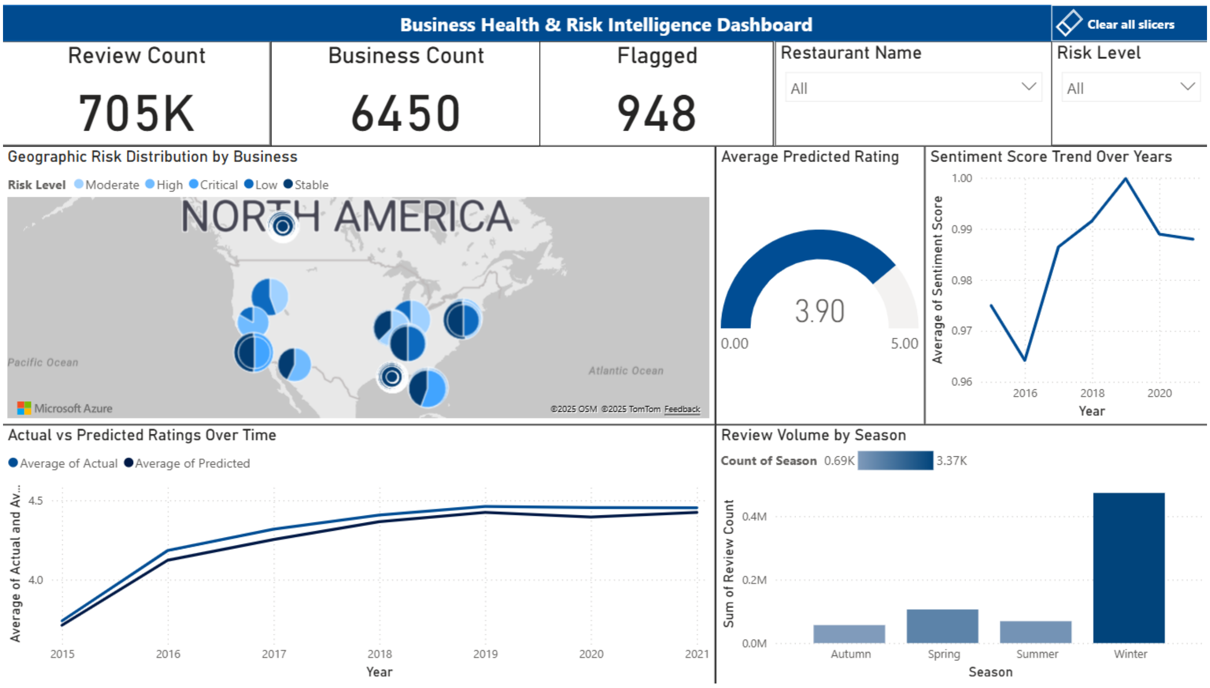
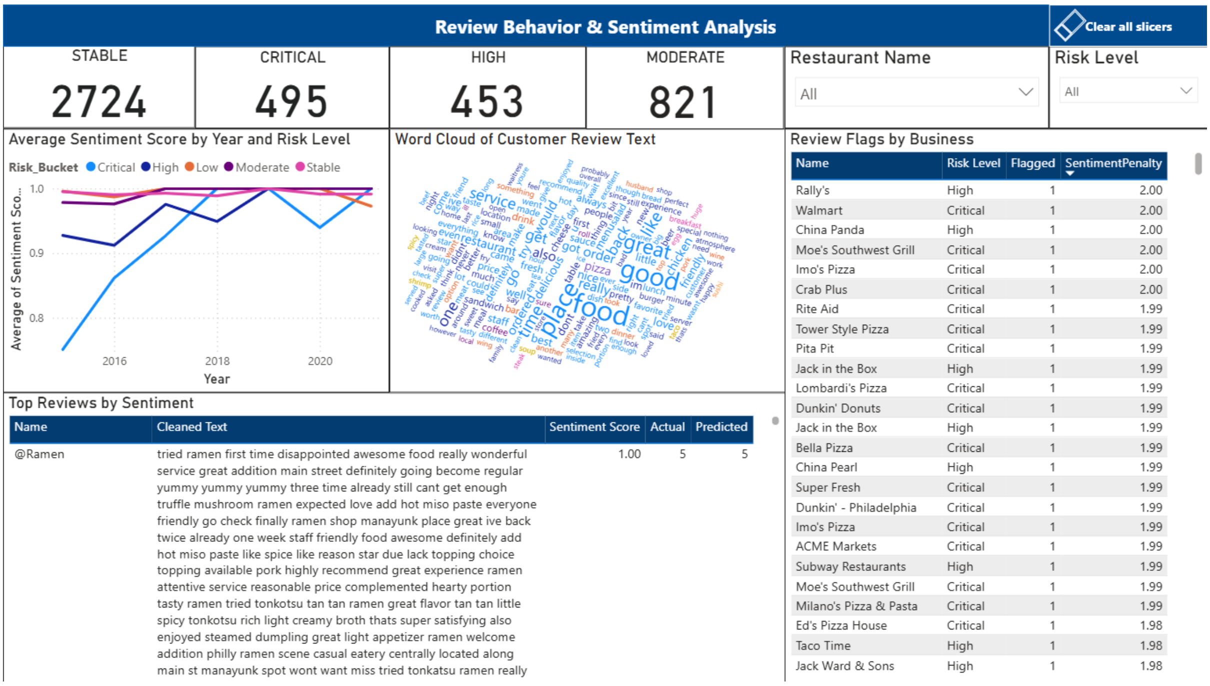


Figure 12: Business Health & Risk Intelligence Dashboard

To contextualize predictions, the dashboard visualizes sentiment trends derived from early reviews using VADER scores (Hutto & Gilbert, 2014). These smoothed trajectories reveal emotional tone shifts such as abrupt sentiment drops that can act as early warning signals (BrightLocal, 2020).

A rule-based risk alert system flags businesses likely to experience declining ratings based on key indicators like low sentiment or review-check-in anomalies. These heuristics complement the model output with interpretable triggers tailored for fast, actionable decisions (Khalid et al., 2023).

A standout feature is the What-If simulator, allowing users to adjust key inputs, sentiment score, review length, vote counts, and observe their effect on predicted outcomes in real time. This not only increases user engagement but also demystifies the model's logic by linking inputs directly to predictions via SHAP values.



Name	Risk Level	Flagged	SentimentPenalty
Rally's	High	1	2.00
Walmart	Critical	1	2.00
China Panda	High	1	2.00
Moe's Southwest Grill	Critical	1	2.00
Imo's Pizza	Critical	1	2.00
Crab Plus	Critical	1	2.00
Rite Aid	Critical	1	1.99
Tower Style Pizza	Critical	1	1.99
Pita Pit	Critical	1	1.99
Jack in the Box	High	1	1.99
Lombardi's Pizza	Critical	1	1.99
Dunkin' Donuts	Critical	1	1.99
Jack in the Box	High	1	1.99
Bella Pizza	Critical	1	1.99
China Pearl	High	1	1.99
Super Fresh	Critical	1	1.99
Dunkin' - Philadelphia	Critical	1	1.99
Imo's Pizza	Critical	1	1.99
ACME Markets	Critical	1	1.99
Subway Restaurants	High	1	1.99
Moe's Southwest Grill	Critical	1	1.99
Milano's Pizza & Pasta	Critical	1	1.99
Ed's Pizza House	Critical	1	1.98
Taco Time	High	1	1.98
Jack Ward & Sons	High	1	1.98

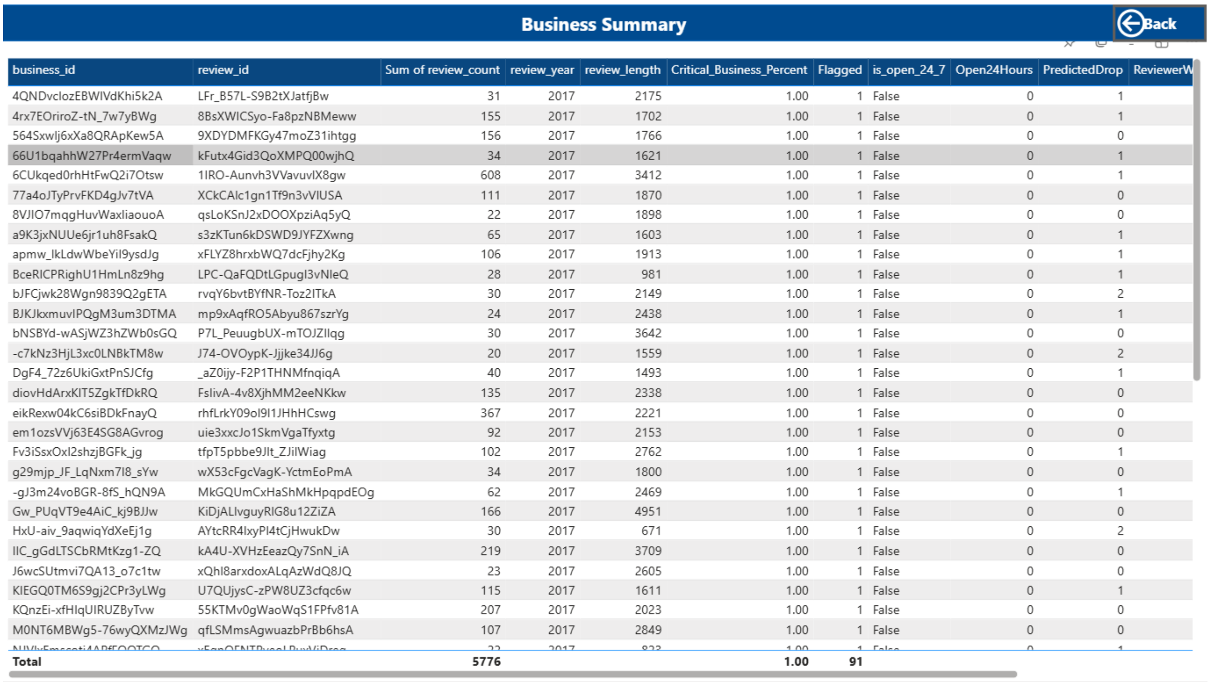
Name	Cleaned Text	Sentiment Score	Actual	Predicted
@Ramen	tried ramen first time disappointed awesome food really wonderful service great addition main street definitely going become regular yummy yummy yummy three time already still cant get enough truffle mushroom ramen expected love add hot miso paste everyone friendly go check finally ramen shop manayunk place great ive back twice already one week staff friendly food awesome definitely add hot miso paste like spice like reason star due lack topping choice topping available pork highly recommend great experience ramen attentive service reasonable price complemented hearty portion tasty ramen tried tonkotsu tan tan ramen great flavor tan tan little spicy tonkotsu rich light creamy broth thats super satisfying also enjoyed steamed dumpling great light appetizer ramen welcome addition philly ramen scene casual eatery centrally located along main st manayunk spot wont want miss tried tonkatsu ramen really	1.00	5	5

Figure 13: Review Behavior & Sentiment Analysis

SHAP explanations are presented as summary bar charts that highlight the most influential features for each prediction. This visual layer enhances transparency, showing users not just what the model predicts, but why.

The dashboard is designed for clarity and accessibility. Color-coded visuals, intuitive filters (e.g., by city, category, hours), and minimal technical jargon make it usable even for non-technical users. Tooltips and contextual help support onboarding and exploration.

Several delivery platforms were evaluated, including Streamlit for custom web apps, Tableau for rapid prototyping, and Excel for low-tech deployment. Power BI was ultimately chosen for its strong balance between interactivity, scalability, and user familiarity. Unlike Streamlit, which requires a custom deployment pipeline and Python literacy, Power BI offers interactivity with advanced visuals, drillthroughs, slicers, and geographic mapping all critical for actionable storytelling. Its seamless integration with enterprise environments and low barrier to entry made it ideal for our SME-focused target users.



Business Summary

Back

business_id	review_id	Sum of review_count	review_year	review_length	Critical_Business_Percent	Flagged	is_open_24_7	Open24Hours	PredictedDrop	ReviewerW
4QNDvclozEBWIVdKhi5k2A	LFr_B57L-S9B2tXJatfjBw	31	2017	2175	1.00	1	False	0	1	
4rx7EOriroZ-tN_7w7yBWg	8BsXWICSyo-Fa8pzNBMeww	155	2017	1702	1.00	1	False	0	1	
564Sxwlj6xXa8QRApKew5A	9XDYDMFKGy47moZ31ihtgg	156	2017	1766	1.00	1	False	0	0	
66U1bqahhW27Pr4ermVaqw	kFutx4Gid3QoXMPQ00wjhQ	34	2017	1621	1.00	1	False	0	1	
6CUkqed0rhHtFwQ2i7Otsw	1IRO-Aunvh3VVavuvlX8gw	608	2017	3412	1.00	1	False	0	1	
77a4oJTyPrvFKD4gJv7tVA	XCkCAlc1gn1Tf9n3vVlUSA	111	2017	1870	1.00	1	False	0	0	
8VJIO7mqgHuvWaxliaouoA	qsLoKSnJ2xDOOXpziAq5yQ	22	2017	1898	1.00	1	False	0	0	
a9K3jxNUUe6jr1uh8FsakQ	s3zKTun6kDSWD9JYFZXwng	65	2017	1603	1.00	1	False	0	1	
apmw_lkLdwWbeYil9ysdJg	xFLYZ8hrxbWQ7dcFjhy2Kg	106	2017	1913	1.00	1	False	0	1	
BceRICPRighU1HmLn8z9hg	LPC-QaFQDtLGpugl3vNleQ	28	2017	981	1.00	1	False	0	1	
bJFCjwk28Wgn9839Q2gETA	rvqY6bvtBYfNR-Toz2ITkA	30	2017	2149	1.00	1	False	0	2	
BJKJkxmuvIPQgM3um3DTMA	mp9xAqfRO5Abyu867szrYg	24	2017	2438	1.00	1	False	0	1	
bNSBYd-wASjWZ3hZWb0sGQ	P7L_PeuugbUX-mTOJZIlqg	30	2017	3642	1.00	1	False	0	0	
-c7kNz3HjL3xc0LNBkTM8w	J74-OVOypK-Jjjke34JJ6g	20	2017	1559	1.00	1	False	0	2	
DgF4_72z6UkiGxtPnSJCfg	_aZ0ijy-F2P1THNMfnqiqA	40	2017	1493	1.00	1	False	0	1	
diovHdArxKIT5ZgkTfDkRQ	FslivA-4v8XjhMM2eeNKkw	135	2017	2338	1.00	1	False	0	0	
eikRexw04kC6siBDkFnayQ	rhfLrkY09ol9I1JHhHCswg	367	2017	2221	1.00	1	False	0	0	
em1ozsVVj63E4SG8AGvrog	uie3xxcJo1SkmVgaTfyxtg	92	2017	2153	1.00	1	False	0	0	
Fv3iSsxOxl2shzjBGFk_jg	tfpT5pbbe9Jlt_ZJilWiag	102	2017	2762	1.00	1	False	0	1	
g29mjp_JF_LqNxm7I8_sYw	wX53cFgcVagK-YctmEoPmA	34	2017	1800	1.00	1	False	0	0	
-gJ3m24voBGR-8fS_hQN9A	MkGQUmCxHaShMkHpqpdEOg	62	2017	2469	1.00	1	False	0	1	
Gw_PUqVT9e4AiC_kj9BJJw	KiDjALlvguyRIG8u12ZiZA	166	2017	4951	1.00	1	False	0	0	
HxU-aiv_9aqwiqYdXeEj1g	AYtcRR4IxyPI4tCjHwukDw	30	2017	671	1.00	1	False	0	2	
IIC_gGdLTSCbRMtKzg1-ZQ	kA4U-XVHzEeazQy7SnN_iA	219	2017	3709	1.00	1	False	0	0	
J6wcSUtmvi7QA13_o7c1tw	xQhI8arxdoxALqAzWdQ8JQ	23	2017	2605	1.00	1	False	0	0	
KIEGQ0TM6S9gj2CPr3yLWg	U7QUjysC-zPW8UZ3cfqc6w	115	2017	1611	1.00	1	False	0	1	
KQnzEi-xfHIqUIRUZByTvw	55KTMv0gWaoWqS1FPfv81A	207	2017	2023	1.00	1	False	0	0	
M0NT6MBWg5-76wyQXMzJWg	qfLSMmsAgwuazbPrBb6hsA	107	2017	2849	1.00	1	False	0	0	
[illegible]	[illegible]	[illegible]	2017	[illegible]	1.00	1	False	0	1	
Total		**5776**			**1.00**	**91**				

Figure 14: Business Summary

By transforming complex model outputs into actionable visual intelligence, the dashboard closes the loop on the data value chain. It empowers users to not only interpret early sentiment signals but also simulate improvements making predictive AI both practical and impactful for real-world SME environments.

3.4.1 Interactive Power BI Dashboard

Access the interactive Power BI dashboard here: Power BI Dashboard

(Includes predicted ratings, sentiment trends, SHAP explanations, and What-If simulations.)

4. Technology Architecture

The artefact's architecture was designed to be modular, scalable, and aligned with the Data Value Map (DVM), ensuring a seamless flow from data ingestion to model delivery. Emphasizing open-source tools and lightweight frameworks, the system remains cost-effective, adaptable, and accessible for small and medium enterprises (SMEs).

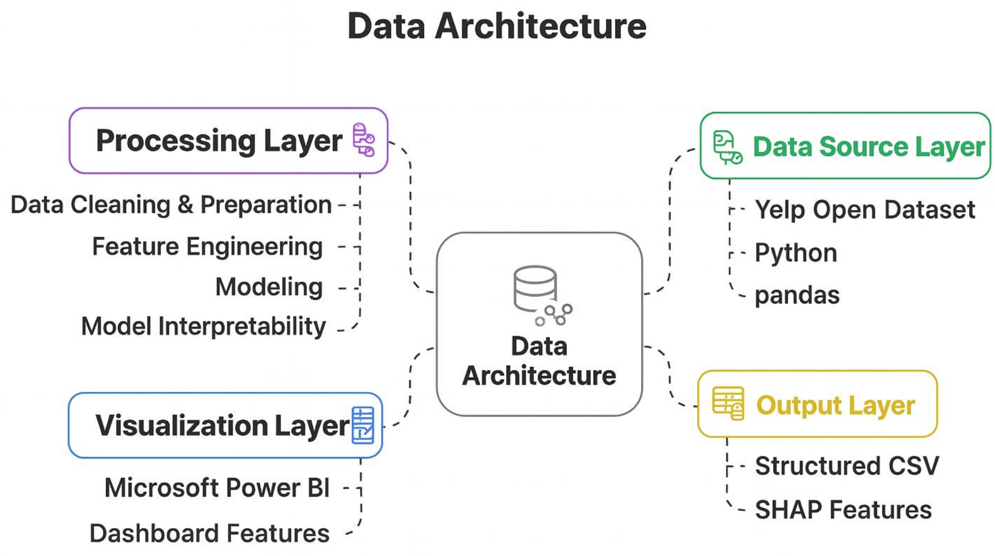


Figure 15: Data Architecture

The architecture begins with Python-driven data acquisition. Yelp's raw *review.json* and *business.json* files were loaded into Pandas DataFrames. Initial filters narrowed the focus to mid-sized food businesses, while nested fields and inconsistent formats were handled using Python's ast, datetime, and custom parsers.

In the integration stage, datasets were merged via business_id. Features were cleaned, type-cast, and label-encoded. New fields such as generalized categories and sentiment scores were engineered to enhance model input quality. Threshold filtering ensured only complete, high-value records were retained (Mishra & Rastogi, 2019).

The analysis layer applied Natural Language Processing using NLTK for tokenization, lemmatization, and part-of-speech tagging. Sentiment was computed using VADER (Hutto & Gilbert, 2014). Cleaned texts were vectorized via TF-IDF and combined with structured features like review length and vote metrics. Models including Logistic Regression, Random Forest, and XGBoost were trained using Scikit-learn and XGBoost libraries. Model evaluation followed standard multiclass metrics, with XGBoost selected for final deployment based on accuracy and robustness (Khalid et al., 2023).

For interpretability, SHAP (SHapley Additive exPlanations) was integrated to assign feature-level impact scores for each prediction (Lundberg & Lee, 2017). These SHAP values were exported as CSVs and later embedded in the dashboard, enabling local and global model explanations.

The delivery layer was built using Power BI. Predictions, SHAP values, and feature inputs were loaded through built-in connectors. The dashboard includes views for predicted vs. actual ratings, sentiment trends, risk alerts, and an interactive What-If simulator. Power BI's dynamic filtering and calculated fields allow users to explore insights by city, category, or business profile.

Together, these components operationalize the four pillars of the DVM: structured acquisition, reliable integration, rigorous analysis, and transparent delivery (Sammon, 2024). Python provides the backend processing power, while Power BI delivers insights in an accessible and actionable format bridging technical rigor with stakeholder usability.

5. Model Evaluation and SHAP Interpretation

Model performance was assessed using standard multi-class classification metrics: accuracy, precision, recall, and F1 score. An 80:20 train-test split with stratification ensured consistent class representation across business star ratings.

Logistic Regression was used as a baseline for its speed and simplicity. After applying feature scaling and class balancing, it achieved 56% accuracy, performing best on extreme classes (1-star and 5-star) but struggling with middle-range predictions due to ambiguous linguistic patterns.

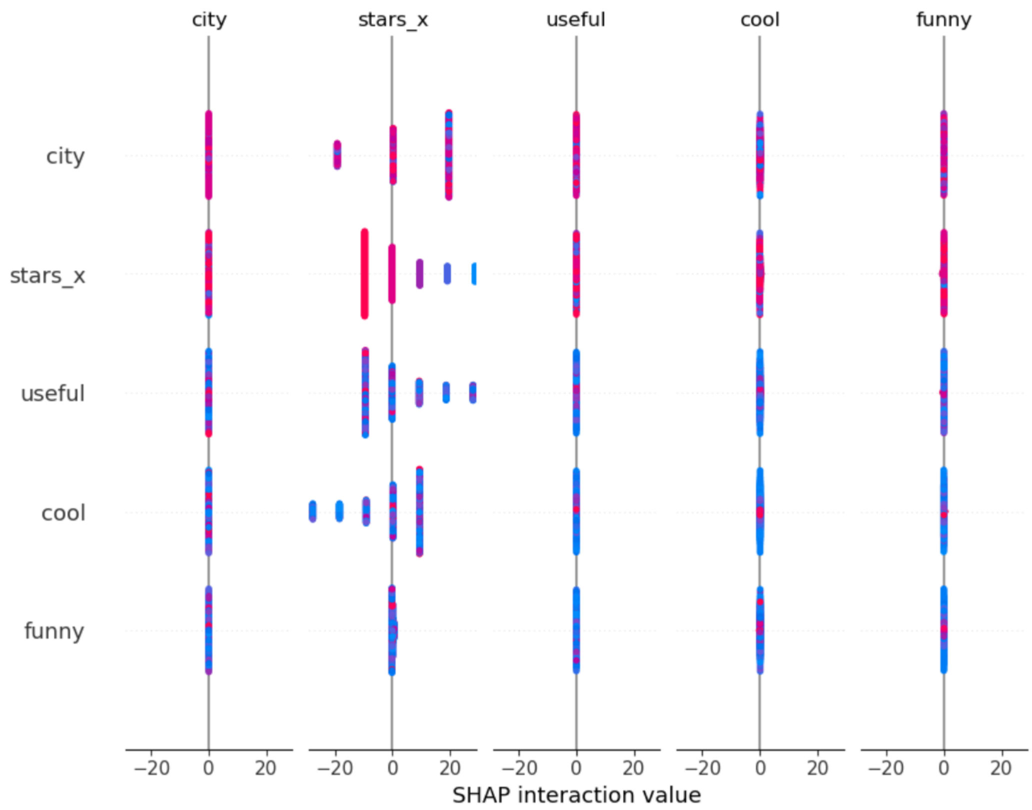


Figure 16: SHAP Interaction Value

Random Forest improved upon this, reaching 57% accuracy. It captured non-linear relationships more effectively, with better macro F1 scores and generalization across 2- to 4-star classes. It also contributed to reliable feature importance analysis while mitigating overfitting.

XGBoost, the best-performing model, achieved 61% accuracy by optimizing a multi-class log loss function and handling both structured and unstructured features efficiently. It showed the strongest balance across precision, recall, and F1 scores and was ultimately selected for deployment in the dashboard.

To ensure transparency, SHAP (SHapley Additive exPlanations) was used to interpret model outputs. SHAP assigns per-feature contributions to each prediction, offering both global insights and local justifications. Key features influencing predictions included sentiment score (via VADER), review length, and engagement metrics such as “useful,” “funny,” and “cool” votes. Grammatical richness (e.g., adjective/adverb counts) and business attributes like category, city, and state also shaped outcomes.

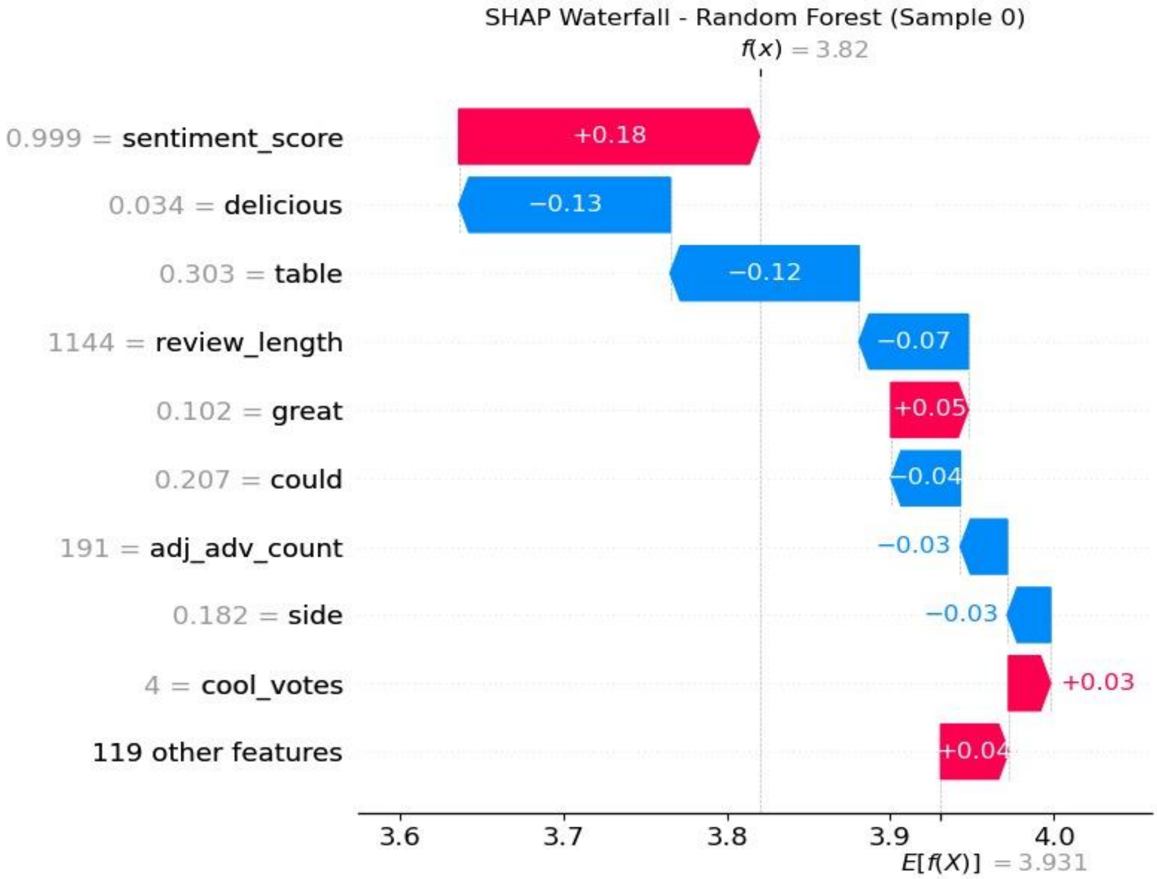


Figure 17: Shap Waterfall - Random Forest (Sample 0)

These SHAP results were visualized within the Power BI dashboard through summary and force plots. Users could inspect which features increased or decreased a given rating, enhancing the artefact's credibility and practical relevance (Lundberg & Lee, 2017; Sammon, 2024). For example, low engagement on otherwise positive reviews signaled muted customer enthusiasm, while short, neutral reviews indicated possible reputational risk (Khalid et al., 2023).

In summary, evaluation and SHAP interpretation validated the artefact both quantitatively and qualitatively — ensuring predictions were not only accurate, but also interpretable and actionable for stakeholders.

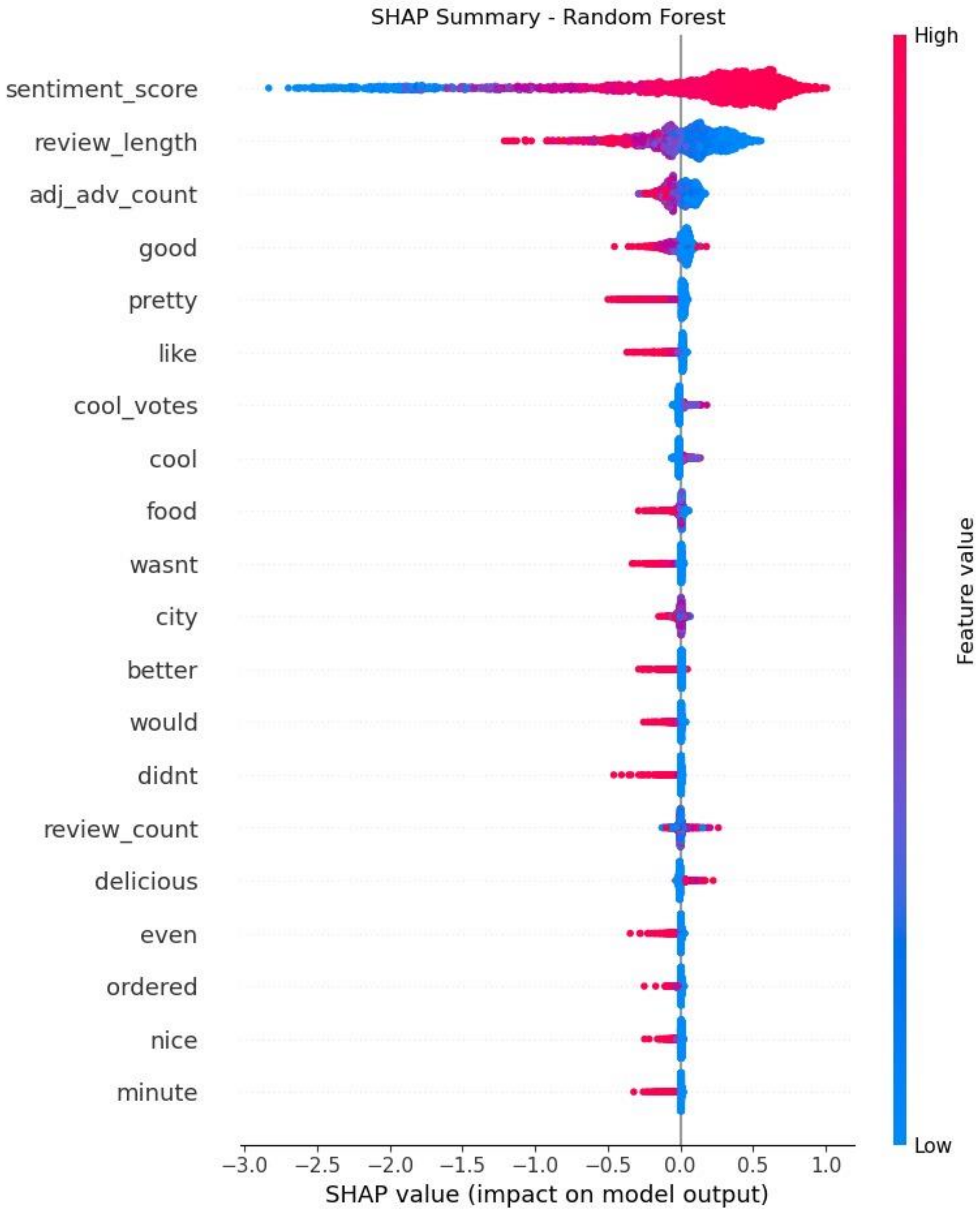


Figure 18: Shap Summary - Random Forest

6. Business Model Canvas

To explore the real-world viability of the artefact, we developed a Business Model Canvas (BMC) that outlines how the system could function as a market-ready product. The core value proposition is proactive churn prevention for small and medium-sized food businesses, using early customer reviews to anticipate long-term reputation risks (BrightLocal, 2020; Luca, 2016). Our target customer segment includes SMBs such as cafés, restaurants, and salons that rely heavily on online platforms for visibility and customer acquisition (Mishra & Rastogi, 2019).

The solution offers a self-service experience through a web dashboard with predictive insights, SHAP-driven explanations, and automated alerts delivered via mobile or email (Lundberg & Lee, 2017). Key activities include data ingestion, machine learning model execution, alert generation, and dashboard maintenance — powered by critical resources like the Yelp dataset, an XGBoost classification engine, and Power BI for visualization (Khalid et al., 2023).

The business would rely on partnerships with Yelp's API and local business networks for onboarding and scale. The cost structure would primarily involve cloud hosting and ongoing development, while revenue would follow a freemium model — with basic features available for free and advanced analytics offered via subscription. Overall, the BMC reflects both the technical potential and commercial feasibility of the artefact.

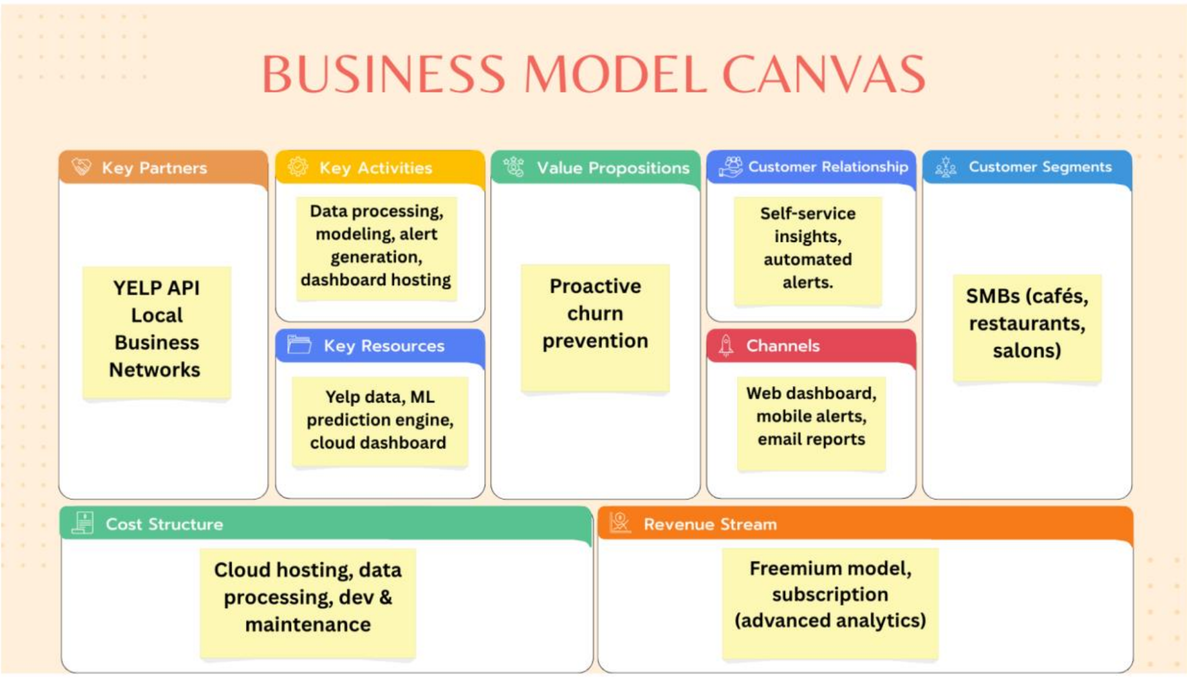


Figure 19: Business Canvas Model

7. Discussion and Reflection

The development of this artefact was an iterative, collaborative, and research-driven effort that blended technical skill, business understanding, and user-centered design. From initial problem framing to the final dashboard delivery, the team adopted a flexible yet structured workflow, balancing analytical rigor with practical usability to bridge the gap between academic machine learning and applied decision-making.

Teamwork was a cornerstone of the project's success. Each member focused on distinct components — data cleaning, feature engineering, model development, SHAP interpretation, or dashboard design. Coordination was maintained through regular meetings, GitHub version control, shared notebooks, and collaborative writing tools. This approach ensured efficient task division while fostering a unified understanding of the artefact's goals.

Key challenges emerged throughout. Early hurdles involved processing Yelp's nested and inconsistent JSON structures, particularly in fields like operating hours and attributes (Yelp, 2023). These were resolved using Python's parsing libraries and custom flattening functions. Another major issue was class imbalance after filtering for food SMEs. This was addressed through weighted classification strategies and ensemble models like Random Forest and XGBoost, which naturally handle uneven distributions better (Khalid et al., 2023; Mishra & Rastogi, 2019).

Due to the original Yelp dataset's large size — over 6.7 million reviews — several preprocessing operations became infeasible on standard personal machines. Some lines of code, such as nested JSON parsing and feature extraction, took hours to run or failed due to memory constraints. To ensure efficient development and avoid system crashes, we strategically reduced the dataset by filtering for relevant business types, removing chains, and limiting the review timeframe (Mishra & Rastogi, 2019; BrightLocal, 2020).

Feature engineering was particularly complex, especially in quantifying subjective traits like authenticity or expressiveness. Variables such as review length, adjective density, and engagement votes were extensively explored and validated via SHAP (Lundberg & Lee, 2017).

These insights confirmed that no single metric sufficed — reliable predictions required a composite of linguistic, behavioral, and contextual indicators (Khalid et al., 2023).

The project also deepened our understanding of applied NLP, multi-class classification, and explainability. Working with SHAP not only clarified model behavior but also enhanced user trust by enabling transparent, case-specific interpretation (Lundberg & Lee, 2017). Designing the What-If simulator in Power BI pushed the team to think from the user's perspective — prioritizing interactivity, clarity, and actionable insight.

The artefact evolved significantly over three development cycles. The first version focused on problem scoping and preliminary cleaning. The second introduced a working prototype with Logistic Regression and basic visualizations. The final iteration delivered a robust, interpretable, and interactive system with advanced modeling and a fully functional Power BI dashboard. This progression reflected the team's growth and adaptability in response to feedback.

In sum, the project was a comprehensive learning journey that integrated academic foundations with real-world application. It sharpened technical competencies, strengthened collaboration, and demonstrated how data science can generate meaningful value for business users. Lessons from this process — especially in data handling, model transparency, and user experience — will continue to guide future work in analytics and system design.

8. Conclusion and Future Enhancements

This project addressed a key challenge faced by small and mid-sized food businesses: the inability to act on customer sentiment before reputational damage occurs (BrightLocal, 2020; Luca, 2016). By building a predictive artefact that estimates long-term star ratings using only early customer reviews, the solution transforms review monitoring from a reactive process into a proactive strategic tool. Through structured preprocessing, explainable machine learning, and stakeholder-centred delivery, the artefact offers practical value and operational foresight (Khalid et al., 2023; Lundberg & Lee, 2017).

The full pipeline adheres to the four stages of the Data Value Map (Sammon, 2024). Data was acquired from the Yelp Open Dataset and filtered to focus on relevant SME businesses (Yelp, 2023). Integration involved cleaning, merging, and feature engineering to produce a structured dataset. Analysis was conducted using multiple classifiers, with XGBoost emerging as the most accurate. Interpretability was achieved through SHAP (Lundberg & Lee, 2017), and insights were delivered via a Power BI dashboard featuring sentiment tracking, risk alerts, and a What-If simulator (Sammon, 2024).

The artefact also aligns with broader development goals. It advances **SDG 8** by helping businesses improve service quality, retain customers, and foster economic resilience. It supports **SDG 9** by embedding AI and interpretability tools into SME infrastructure in a scalable and ethical manner (United Nations, n.d.). The project demonstrates that advanced analytics can be both accessible and impactful, contributing to a wider vision of responsible AI adoption.

Looking ahead, several enhancements could strengthen the system. Transitioning to real-time prediction via APIs would enable dynamic updates as new reviews arrive, increasing responsiveness in fast-paced environments. Expanding language support would allow broader adoption in multilingual markets. Integrating other review platforms such as Google Reviews or TripAdvisor would offer a more comprehensive view of public sentiment.

On the modeling front, incorporating advanced architectures such as fine-tuned BERT could capture deeper semantic patterns and further improve accuracy (OpenAI, 2023). The dashboard could also benefit from personalization features that allow filtering by customer demographics or business type. Additionally, pairing SHAP with other explanation tools like LIME could enhance interpretability by offering diverse perspectives on model behavior (Lundberg & Lee, 2017).

In conclusion, this artefact proves that early review data can be leveraged to forecast long-term business outcomes in a way that is data-driven, explainable, and immediately actionable. It provides a robust foundation for future research, commercial scaling, and cross-platform integration while supporting the larger goals of digital innovation and economic empowerment.

9. References

- BrightLocal. (2020). *Local consumer review survey*. Retrieved from https://www.brightlocal.com/research/local-consumer-review-survey/
- Devlin, J., Chang, M-W., Lee, K., & Toutanova, K. (2018). *BERT: Pre-training of deep bidirectional transformers for language understanding*. arXiv. Retrieved from https://arxiv.org/abs/1810.04805
- Hutto, C. J., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the International AAAI Conference on Web and Social Media, 8*(1), 216–225. Retrieved from https://ojs.aaai.org/index.php/ICWSM/article/view/14550
- Khalid, A., Ahmad, A., Raza, H., & Aslam, M. (2023). *Improved restaurant review analysis using VADER-based sentiment analysis and automatic rating matching* [Preprint]. ResearchGate. Retrieved from https://www.researchgate.net/publication/372404011
- Luca, M. (2016). *Reviews, reputation, and revenue: The case of Yelp.com* (Harvard Business School Working Paper No. 12-016). Retrieved from https://www.hbs.edu/faculty/Pages/item.aspx?num=41233
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems, 30*, 4765–4774. Retrieved from https://proceedings.neurips.cc/paper_files/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf
- Mishra, A., & Rastogi, R. (2019). Predicting restaurant ratings using review text and sentiment analysis. In *Proceedings of ICACCCN 2019* (pp. 1–6). IEEE. https://doi.org/10.1109/ICACCCN.2019.8904652
- Sammon, D. (2024). The Data Value Map unpacked. *Journal of Need, 10*(1), 1–4.
- TextBlob Developers. (n.d.). *TextBlob: Simplified text processing*. Retrieved from https://textblob.readthedocs.io/en/dev/
- United Nations. (n.d.). *Sustainable Development Goals*. Retrieved from https://sdgs.un.org/goals

- Yelp. (2023). *Yelp Open Dataset documentation*. Retrieved from https://www.yelp.com/dataset